

ASYMPTOTICALLY OPTIMAL TESTS OF POWER ONE FOR THE LOCATION PARAMETER OF AN EXPONENTIAL DISTRIBUTION

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SUMMARY

Tests with power one were introduced by Robbins (see e.g. 1970, 1974) and developed mainly in the case of exponential families. In this article we consider such tests for an exponential distribution with unknown location parameter θ . We use methods of Farrell (1964) and Robbins and Siegmund (1974) to prove that one may restrict oneself to certain one-sided procedures based on sequences of such procedures. Finally we construct a procedure for which this lower bound is achieved, and this procedure is therefore in a sense asymptotically optimal.

1. Tests of power one and reduction to sample minima

Suppose the lifetime of an electronic device has an exponential distribution

$$f_{\theta}(x) = \exp \{-(x - \theta)\} I(x \geq \theta)$$

where $I(A)$ denotes the indicator function of the event A . The device is manufactured according to a standard method and experience has shown that $\theta = \theta_0$. A new method for manufacturing the device comes in use alongside the standard method and items are produced simultaneously, using both methods. The question arises whether the new method yields a higher lifetime, i.e. whether $\theta > \theta_0$. If this is so, it should be discovered as soon as possible and with certainty. If this need not be so, there is no objection against letting both methods continue. This means we wish to test, for the new method,

$$H_0 : \theta = \theta_0 \text{ against } H_1 : \theta > \theta_0$$

in such a way that

$$P_{\theta_0}(\text{reject } H_0) \leq \alpha, \text{ where } \alpha > 0 \text{ is prescribed and}$$

$$P_{\theta}(\text{reject } H_0) = 1 \text{ for } \theta > \theta_0.$$

Also, if N denotes the number of observations taken, $E_\theta N$ should be as small as possible, for $\theta > \theta_0$.

Assume $\theta_0 = 0$ without loss of generality. We restrict ourselves to tests T of power one, rejecting H_0 if and only if $N_T < \infty$ (where N_T denotes the stopping variable of T) and satisfying $P_\theta(N_T < \infty) = 1$ for $\theta > 0$ and $0 < P_0(N_T < \infty) < 1$.

The development in this section is analogous to that in Farrell (1964) (section 3) and the notation is similar.

Let $\{X_n, n \geq 1\}$ be a sequence of independent observations with common density function $f_\theta(x)$. For purposes of randomization, introduce a sequence of independent random variables $\{Z_{nT}, n \geq 0\}$ associated with a test T . The sequence $\{Z_{nT}\}$ is assumed to be independent of $\{X_n\}$ and the distribution of $\{Z_{nT}\}$ does not depend on θ .

A test T is performed by first observing Z_{0T} and then sequentially the pairs $(X_1, Z_{1T}), (X_2, Z_{2T}), \dots$. For testing H_0 against H_1 let $T: A(n)$ denote the rejection region and $T: C(n)$ the continuation region after observing $V_{nT} = (Z_{0T}, X_1, Z_{1T}, \dots, X_n, Z_{nT})$, $n \geq 0$. A realization of V_{nT} is denoted by $v_{nT} = (z_{0T}, x_1, z_{1T}, \dots, x_n, z_{nT})$.

Let $M_n = \min(X_1, \dots, X_n)$ and $m_n = \min(x_1, \dots, x_n)$, $n = 1, 2, \dots$. Recall that for each n the statistic M_n is sufficient for θ . Hence the main result of this section as stated in Theorem 1.1 is not unexpected.

Definition: The test S is a test with upper boundary $\{b_n, n \geq 1\}$ if $v_{nS} \in S: A(n)$ is equivalent to $m_n \geq b_n$, $n = 1, 2, \dots$.

Theorem 1.1

Let T be any test of power one for H_0 against H_1 such that

- (i) $P_\theta(N_T < \infty) = 1$ for $\theta > 0$
- (ii) $P_0(N_T < \infty) = \alpha_T$ where $0 < \alpha_T < 1$.

Then there exists a test S of power one with upper boundary such that

- (a) $P_\theta(N_S \geq n) \leq P_\theta(N_T \geq n)$ for $n = 1, 2, \dots$ and $\theta > 0$ which implies $E_\theta(N_S) \leq E_\theta(N_T)$, $\theta > 0$.
- (b) $\alpha_S = P_0(N_S < \infty) = \alpha_T$.

The proof of the theorem is based on the following lemma:

Lemma 1.1

Let T be any test of power one satisfying (i) and (ii) of Theorem 1.1. Then there exists a sequence $\{T_n, n = 0, 1, \dots\}$ of tests with the following properties:

$$(i) \quad T_0 = T$$

$$(ii) \quad T_m: A(j) = T_n: A(j) \text{ for } 0 \leq m \leq n, 0 \leq j \leq m$$

$$(iii) \quad P_\theta(N_{T_n} \geq m) \geq P_\theta(N_{T_{n+1}} \geq m) \text{ for } \theta > 0, m \geq 0, n \geq 0$$

$$(iv) \quad P_0(V_m \in T_{n+1}: A(m)) = P_0(V_m \in T_n: A(m)) \text{ for } m \geq 0, n \geq 0$$

$$(v) \quad Z_{mT_n} = Z_{mT_{n-1}}, m \geq 0, n \geq 1 \\ = Z_m(szy)$$

$$(vi) \quad v_n \in T_n: A(n) \text{ is equivalent to } m_n \geq b_n \text{ for some sequence } \{b_n\}, n \geq 1.$$

Proof of Theorem 1.1 from Lemma 1.1

Define the test S by $S: A(m) = T_m: A(m)$ for each $m \geq 0$ and let $N_S = m$ if and only if $v_{mS} \in S: A(m)$, i.e. S rejects H_0 if and only if $N_S < \infty$. Here $Z_{mS} = Z_m, m \geq 0$. Then (a) follows using (i), (ii) and (iii) of Lemma 1.1, while (b) follows from (i) and (iv). Also (vi) of Lemma 1.1 implies that S is a test with upper boundary and part (a) of the theorem shows that S is a test of power one.

The following lemma, the proof of which is immediate, will be used repeatedly in the remainder of this paper.

Lemma 1.2

Let B_θ be an event depending on $X_1, \dots, X_n$ and on θ . Then

$$P_\theta(B_\theta) \leq \exp \{n\theta\} P_0(B_\theta) \text{ for all } \theta > 0.$$

The reverse inequality holds for $\theta < 0$.

Proof of Lemma 1.1

Let $n \geq 1$ be fixed and suppose T_{n-1} has been defined. We now construct T_n . Write $V_{mT_n} = V_m$, etc.

For $m \leq n-1$, define $T_n: A(m) = T_{n-1}: A(m)$ and $T_n: C(m) = T_{n-1}: C(m)$.

For $m = n$, let b_n be some number which will be determined later on in the proof and consider the sets

$$E_1 = \{v_n: m_n \geq b_n, v_n \in T_{n-1}: C(n)\}$$

$$E_2 = \{v_n: m_n < b_n, v_n \in T_{n-1}: A(n)\}.$$

Now define

$$T_n: A(n) = E_1 \cup [T_{n-1}: A(n) - E_2] = \{m_n \geq b_n\} \cap \{N_{T_{n-1}} \geq n\}$$

$$T_n: C(n) = E_2 \cup [T_{n-1}: C(n) - E_1] = \{m_n < b_n\} \cap \{N_{T_{n-1}} \geq n\}$$

For $m \geq n + 1$, let W_n be a $(2n + 1)$ -dimensional random vector taking values in E_1 a.s. P_0 and defined by $Z_n = \phi_{2n+1}(W_n)$ where ϕ_k is a 1-1 onto measurable function such that ϕ_k^{-1} is also measurable, $k \geq 1$. Define

$$T_n: A(m) = \{v_m: v_n \in T_{n-1}: C(n) - E_1, v_m \in T_{n-1}: A(m)\} \cup$$

$$\{v_m: v_n \in E_2, (w_n, x_{n+1}, z_{n+1}, \dots, x_m, z_m) \in T_{n-1}: A(m)\}$$

$$T_n: C(m) = \{v_m: v_n \in T_{n-1}: C(n) - E_1, v_m \in T_{n-1}: C(m)\} \cup$$

$$\{v_m: v_n \in E_2, (w_n, x_{n+1}, z_{n+1}, \dots, x_m, z_m) \in T_{n-1}: C(m)\}.$$

Note that the test T_n is constructed from T_{n-1} in such a way that (vi) is satisfied for T_n (case $m = n$ above) and this forces us to define T_n at stages $m \geq n + 1$ in a way such as the above.

We now show that, except for a special case, b_n may be defined such that $P_0(V_n \in E_1) = P_0(V_n \in E_2)$, an equality which plays an important role in the remainder of the proof.

Let

$$g_1(r) = P_0(M_n \geq r, V_n \in T_{n-1}: C(n))$$

$$g_2(r) = P_0(M_n < r, V_n \in T_{n-1}: A(n))$$

Then g_1 and g_2 are continuous, g_1 is non-increasing and g_2 is non-decreasing. Also

$g_1(r) \rightarrow 0$ as $r \rightarrow \infty$, while $g_2(0) = 0$. Consider the following cases:

Case 1: There exists an $r > 0$ such that $g_1(r) > 0$ and $g_2(r) > 0$.

(1.1) Define b_n to be the unique solution of $g_1(r) = g_2(r)$. Then

$$0 < b_n < \infty.$$

Case 2: There exists an $r > 0$ such that $g_1(r) > 0$. If $r > 0$, then

$$g_2(r) = 0.$$

Now there is no solution for (1.1) and we define $b_n = +\infty$ with the convention that in this case $\{v_n: m_n \geq b_n\} = \phi$.

Case 3: $g_1(r) = 0$ for all $r > 0$. Define $b_n = 0$.

The proof of (iii) and (iv) will be given separately for each of the above cases. First note:

$$\text{If } m \geq n + 1: P_\theta(V_m \in D) - P_\theta(V_m \in D')$$

$$(1.2) = P_\theta(V_n \in E_2) P_\theta((W_n, \dots, X_m) \in D') - P_\theta(V_n \in E_1, V_m \in D')$$

where D and D' are either $T_n: A(m)$ and $T_{n-1}: A(m)$ or $T_n: C(m)$ and $T_{n-1}: C(m)$. If $m = n$: $P_\theta(V_n \in T_n: C(n)) - P_\theta(V_n \in T_{n-1}: C(n))$

$$(1.3) = P_\theta(V_n \in E_2) - P_\theta(V_n \in E_1)$$

$$\text{while } P_\theta(V_n \in T_n: A(n)) - P_\theta(V_n \in T_{n-1}: A(n))$$

$$(1.4) = P_\theta(V_n \in E_1) - P_\theta(V_n \in E_2)$$

Case 2: Here we have $P_0(V_n \in E_2) = P_0(V_n \in E_1) = 0$ since $g_2(r) = 0$ and $E_1 = \phi$. Using Lemma 1.2 this implies that all the differences in (1.2), (1.3) and (1.4) are zero for $\theta \geq 0$.

Case 3: The proof is completely analogous to that for case 2.

Case 1: First note that the difference in (1.4) is zero for $\theta = 0$ by definition of b_n . Now let $\theta > 0$ and consider (1.3). If $b_n \geq \theta$

$$\begin{aligned}
& P_0(V_n \in E_2) \\
&= P_0(V_n \in E_2, M_n < \theta) + \int_{E_2 \cap \{m_n \geq \theta\}} \exp \{-n\theta\} \pi_{i=1}^n f_\theta(x_i) dx_i \pi_{i=0}^n h_i(z_i) dz_i \\
&= P_0(V_n \in E_2; M_n < \theta) + e^{-n\theta} P_\theta(V_n \in E_2)
\end{aligned}$$

and in the same way

$$P_0(V_n \in E_1) = \exp \{-n\theta\} P_\theta(V_n \in E_1)$$

Hence if $b_n \geq \theta$, the difference in (1.3) becomes

$$-\exp \{n\theta\} P_0(V_n \in E_2, M_n < \theta) \leq 0.$$

On the other hand, if $b_n < \theta$ we have $P_\theta(V_n \in E_2) = 0$ so that the same conclusion holds for the difference in (1.3).

Now proceed to the case where $m \geq n+1$ and consider (1.2). Let W_n have density function $\pi_{i=1}^n f_0(y_i) \pi_{i=0}^n h_i(u_i) / \Delta$ for $w_n = (u_0, y_1, u_1, \dots, y_n, u_n) \in E_1$ and where $\Delta = P_0(V_n \in E_1) > 0$. For $b_n > \theta$ and $\theta \geq 0$ we have

$$\begin{aligned}
& P_\theta((W_n, X_{n+1}, Z_{n+1}, \dots, X_m, Z_m) \in D') \\
&= \int_{E_1 \cap \{m_n \geq \theta\}} \Delta^{-1} P_\theta[(W_n, X_{n+1}, Z_{n+1}, \dots, X_m, Z_m) \in D' \mid W_n = w_n] \\
&\quad \cdot \pi_{i=1}^n f_0(y_i) dy_i \pi_{i=0}^n h_i(u_i) du_i \\
&= \Delta^{-1} \exp \{-n\theta\} \int_{E_1 \cap \{m_n \geq \theta\}} \int_D \pi_{i=1}^m f_\theta(y_i) dy_i \pi_{i=0}^m h_i(u_i) du_i \\
(1.5) \quad &= \Delta^{-1} \exp \{-n\theta\} P_\theta(V_n \in E_1, V_m \in D')
\end{aligned}$$

Now let $\theta = 0$ and substitute (1.5) and $\Delta = P_0(V_n \in E_1) = P_0(V_n \in E_2)$ in (1.2), which proves (iv). If $\theta > 0$, and using Lemma 1.2, we get $\Delta \geq \exp \{-n\theta\} P_\theta(V_n \in E_2)$, which together with (1.5) is now substituted in (1.2), showing that the difference in (1.2) is non-positive.

On the other hand, if $b_n \leq \theta$ one has $P_\theta(V_n \in E_2) = 0$, which gives the desired result for $\theta > 0$ on substitution in (1.2). The case $\theta = 0$ need not be considered now since $b_n > 0$ in case 1.

2. An asymptotic lower bound for the expected sample size of tests of power one

Theorem 2.1

For any test T of power one with stopping variable N_T the following hold:

$$(2.1) \limsup_{\theta \downarrow 0} \theta (\log_2 \theta^{-1})^{-1} E_{\theta} N_T \geq P_0(N_T = \infty)$$

$$(2.2) \lim_{\theta \downarrow 0} \theta E_{\theta} N_T = \infty$$

(We write $\log \log = \log_2$, etc.)

The following lemmas will be used in the proof of Theorem 2.1:

Lemma 2.1

Let n_0 be a positive integer (independent of θ) such that $n_0 < n_1 = [K\theta^{-1}]$ for arbitrary $K > 0$. Also let Q be an arbitrary stopping variable for $X_1, X_2, \dots$. Then

$$(i) \lim_{\theta \downarrow 0} P_{\theta}(Q > n_0) = P_0(Q > n_0)$$

$$(ii) \lim_{n_0 \rightarrow \infty} \limsup_{\theta \downarrow 0} P_{\theta}(n_0 < Q \leq n_1) = 0$$

Proof:

- (i) Let A be an event depending on $X_1, \dots, X_{n_0}$, but not on θ . Using dominated convergence it may be shown that

$$\lim_{\theta \downarrow 0} P_{\theta}(A) = P_0(A).$$

Applying this to $A = \{Q \leq n_0\}$, we obtain (i).

- (ii) This is an immediate consequence of Lemma 1.2.

Lemma 2.2

Let T be a test of power one with upper boundary $\{b_n\}$ and stopping variable N_T . Then

$$(i) \quad b_n \downarrow 0 \text{ as } n \rightarrow \infty$$

$$(ii) \quad nb_n \rightarrow \infty \text{ as } n \rightarrow \infty$$

$$(iii) \quad \limsup_{n \rightarrow \infty} nb_n / \log_2 n \geq 1$$

- (iv) $\lim_{n \rightarrow \infty} P_{\theta_n}(N_T < n) = P_0(N_T < \infty)$ where $\{\theta_n\}$ is any real number sequence, satisfying $0 < \theta_n \leq \rho b_n$ for arbitrary $0 < \rho < 1$.

Proof:

- (i) Since $\{M_n\}$ is a non-increasing sequence, it may be assumed without loss of generality that $b_n \downarrow$. But $b_n \downarrow C > 0$ leads to $P_\theta(N_T < \infty) < 1$ for $\theta < C$.
- (ii) If $n_i b_{n_i} \leq c$ for some sequence of integers $\{n_i\}$ and constant $c > 0$, then
- $$P_0(M_{n_i} \geq b_{n_i}) \geq e^{-c} > 0$$
- which implies $P_0(M_{n_i} \geq b_{n_i} \text{ i.o.}) = 1$, contradicting $P_0(N_T < \infty) < 1$.
- (iii) If, for some ν , $b_n < \log_2 n/n$ for all $n \geq \nu$
- $$P_0(N_T < \infty) \geq P_0(M_n \geq \log_2 n/n \text{ i.o.}) = 1$$
- using (2.5) of Robbins and Siegmund (1972).
- (iv) The proof is analogous to that of Lemma 4, p. 57 of Farrell (1964) and therefore we give only the main steps.

Let $\{\alpha_n\}$ be a real number sequence such that $\alpha_n \downarrow 0$ and $\lim_{n \rightarrow \infty} \alpha_n g(n) = \infty$, where $g(n) = nb_n$. Define $k(n)$ to be the largest integer m such that $b_m/b_n \geq \alpha_m g(m)$. If $b_m/b_n < \alpha_m g(m)$ for all m , define $k(n) = 0$. Then

$$(2.3) \quad \lim_{n \rightarrow \infty} b_{k(n)}/b_n = \infty$$

$$(2.4) \quad \lim_{n \rightarrow \infty} k(n)b_n = \infty$$

are immediate consequences of the definition of $k(n)$. We now prove the following:

$$(2.5) \quad \lim_{n \rightarrow \infty} P_{\rho b_n}(k(n) < N_T \leq n) = 0$$

$$(2.6) \quad \lim_{n \rightarrow \infty} P_{\theta_n}(N_T \leq k(n)) = P_0(N_T < \infty)$$

For $m \leq n$ we have

$$\begin{aligned}
P_{\rho b_n}(N_T = m) &= \int_{\rho b_n}^{\infty} P(N_T = m | M_m = x) m \exp \{-m(x - \rho b_n)\} dx \\
&= \exp \{-(1 - \rho)mb_n\} \int_{b_n}^{\infty} P(N_T = m | M_m = x) m \exp \{-m(x - b_n)\} dx \\
&\quad \text{since } N_T = m \text{ implies } M_m \geq b_m \geq b_n \\
&= \exp \{-(1 - \rho)mb_n\} P_{b_n}(N_T = m)
\end{aligned}$$

Hence $P_{\rho b_n}(k(n) < N_T \leq n) \leq \exp \{-(1 - \rho)k(n)b_n\} \rightarrow 0$, using (2.4).

For (2.6) we first prove that for $\ell \leq n^* \leq n$ and $0 \leq \theta \leq \rho b_n$, the function $P_{\theta}(\ell \leq N_T \leq n^*)$ is non-decreasing and convex in θ . For $0 \leq \theta \leq \rho b_m$ the first and second derivatives of $P_{\theta}(N_T = m)$ are respectively given by $mP_{\theta}(N_T = m)$ and $m^2P_{\theta}(N_T = m)$, which are both non-negative. Now note that $\theta \leq \rho b_n \leq \rho b_{n^*} \leq \rho b_m$ and $P_{\theta}(\ell \leq N_T \leq n^*) = \sum_{m=\ell}^{n^*} P_{\theta}(N_T = m)$. Now (2.6) and (iv) follow in the same way as in Farrell (1964).

Proof of Theorem 2.1:

Theorem 1.1 implies that we may assume T to be a test with upper boundary $\{b_n\}$, say. Then

$$\begin{aligned}
E_{\theta} N_T &\geq n_1 P_{\theta}(N_T > n_1) \\
&= n_1 [P_{\theta}(N_T > n_0) - P_{\theta}(n_0 < N_T \leq n_1)]
\end{aligned}$$

This gives $\liminf_{\theta \downarrow 0} \theta E_{\theta} N_T \geq KP_0(N_T = \infty)$ from (i) and (ii) of Lemma 2.1. Now let $K \uparrow \infty$

to prove (2.2).

For (2.1), using (iii) of Lemma 2.2, let $\{m_i\}$ be a sequence of integers such that

$$m_i b_{m_i} / \log_2 m_i \geq 1 - \epsilon \text{ for all } i \text{ and } \epsilon > 0$$

and $\{\theta_i\}$ a real number sequence such that $0 < \theta_i \leq b_{m_i}$ and $m_i \theta_i / \log_2 m_i = 1 - \epsilon$ for all i . Then $E_{\rho \theta_i} N_T \geq m_i P_{\rho \theta_i}(m_i < N_T < \infty)$ which gives

$$\limsup_{i \rightarrow \infty} \rho \theta_i E_{\rho \theta_i} N_T / \log_2 (\rho \theta_i)^{-1} \geq \rho(1 - \epsilon) P_0(N_T = \infty)$$

using (iv) of Lemma 2.2. Now let $\rho \uparrow 1$ and $\epsilon \downarrow 0$.

3. An asymptotically optimal test

We now construct a test of power one with upper boundary $\{b_n\}$ such that $b_n \sim n^{-1} \log_2 n$ and for which the asymptotic expected sample size as $\theta \downarrow 0$ attains the lower bound in (2.1). We first show that such a sequence $\{b_n\}$ may be found such that $P_0(M_n \geq b_n \text{ for some } n \geq m_j) \leq \alpha$ ($0 < \alpha < 1$) for $j = 3, 4, \dots$ where $m_1 < m_2 < \dots$ is a sequence of positive integers with $m_j \uparrow \infty$ (cf. Darling and Robbins (1967)).

Let $c_n = t_i$ for $m_i \leq n < m_{i+1}$ ($i = 3, 4, \dots$) where

$$t_i = m_i^{-1} (\log i + 2 \log_2 i + \log K_j \alpha^{-1})$$

$$m_i = \lceil \exp \{i / \log i\} \rceil, K_j = \sum_{i=j}^{\infty} [i(\log i)^2]^{-1}.$$

Then

$$\begin{aligned} & P_0(M_n \geq c_n \text{ for some } n \geq m_j) \\ & \leq \sum_{i=j}^{\infty} P_0(M_n \geq t_i \text{ for some } m_i \leq n < m_{i+1}) \\ & = \sum_{i=j}^{\infty} P_0(M_{m_i} \geq t_i) \\ & = \sum_{i=j}^{\infty} \exp \{-m_i t_i\} \\ & = \alpha \end{aligned}$$

From $m_i \leq n < m_{i+1}$ follows $\log i \leq \log_2 n + \log_3 n + \log 2$, $\log_2 i \leq \log_3 n + \log 2$, $m_i > n \exp \{-(\log_2 n)^{-1}\}$, which imply $c_n = t_i < (\log_2 n + 3 \log_3 n + 3 \log 2 + \log K_j \alpha^{-1}) n^{-1} \exp \{(\log_2 n)^{-1}\} = b_n$ (say).

In what follows, N will denote the stopping variable of the above test.

Lemma 3.1

Let $n_1^* = \lceil \theta^{-1} \rceil$ and for arbitrary $0 < \lambda < 1$, let $n_2 = \lceil \lambda \theta^{-1} \log_2 \theta^{-1} \rceil$. Then

$$\lim_{\theta \downarrow 0} P_{\theta}(n_1^* < N \leq n_2) = 0$$

Proof:

Using Lemma 1.2 we obtain

$$\begin{aligned} P_{\theta}(n_1^* < N \leq n_2) & \leq (\log n_1^*)^{\lambda + o(1)} P_0(n_1^* < N \leq n_2) \\ & \leq (\log n_1^*)^{\lambda + o(1)} P_0(M_n \geq n^{-1} \log_2 n \text{ for some } n_1^* < n \leq n_1^* \log_2 n_1^*) \end{aligned}$$

$\rightarrow 0$ if $\theta \downarrow 0$ since

$$\begin{aligned}
 & P_0(M_n \geq n^{-1} \log_2 n \text{ for some } i \leq n \leq i \log_2 i) \\
 & \leq \sum_{k=a}^b P_0(M_n \geq n^{-1} \log_2 n \text{ for some } r_k \leq n \leq r_{k+1}) \\
 & \quad \text{where for } 1 < r < \lambda^{-1}, r_k = \lfloor r^k \rfloor, r_a = i, r_{b+1} = i \log_2 i \\
 & \leq \sum_{k=a}^b P_0(M_{r_k} \geq c_{r_{k+1}}), \text{ where } c_n = n^{-1} \log_2 n \\
 & \leq \sum_{k=a}^b (\log r_k)^{-1/r} \\
 & \leq (b - a + 1)(\log r_a)^{-1/r} \\
 & = \log_3 i / (\log i)^{1/r} \log r \text{ from } a = \log i / \log r \text{ and } b + 1 = (\log i + \log_3 i) / \log r
 \end{aligned}$$

Theorem 3.1

$$\lim_{\theta \downarrow 0} \theta (\log_2 \theta^{-1})^{-1} E_\theta N = P_0(N = \infty)$$

Proof:

$$E_\theta N \geq n_2 [P_\theta(N > n_0) - P_\theta(n_0 < N \leq n_1^*) - P_\theta(n_1^* < N \leq n_2)]$$

If first $\theta \downarrow 0$ and then $n_0 \rightarrow \infty$ using Lemma 2.1 and Lemma 3.1, we obtain

$$\liminf_{\theta \downarrow 0} \theta (\log_2 \theta^{-1})^{-1} E_\theta N \geq P_0(N = \infty)$$

since λ is arbitrary.

For the converse let $n_3 = \lceil \theta^{-1} (\log \theta^{-1})^{-1} \rceil$ and for arbitrary $\lambda > 1$, $n_4 = \lceil \lambda \theta^{-1} \log_2 \theta^{-1} \rceil$. Then

$$\begin{aligned}
 E_\theta N &= \sum_{n=0}^{\infty} P_\theta(N > n) \\
 &\leq n_3 + n_4 P_\theta(N > n_3) + \sum_{n=n_4}^{\infty} P_\theta(N > n) \\
 &\leq n_3 + n_4 \exp \{n_3 \theta\} P_0(N > n_3)
 \end{aligned}$$

using Lemma 1.2 and the fact that $\sum_{n=n_4}^{\infty} P_\theta(N > n) = 0$ for all θ close to 0. This is so,

since $P_\theta(N > n) \leq P_\theta(M_n < b_n) = 1 - \exp \{-(b_n - \theta)n\}$ if $b_n > \theta$ and $= 0$ otherwise. But $b_{n_4} \leq \theta$ for all θ close to 0 (since $\lambda > 1$), implying $b_n \leq b_{n_4} \leq \theta$ for all $n \geq n_4$. Hence

$$\lim_{\theta \downarrow 0} \sup \theta (\log_2 \theta^{-1})^{-1} E_\theta N \leq P_0(N = \infty).$$

Remarks

- (1) Note that the test constructed in Section 3 is also a test of power one of size α for $H_0: \theta \leq 0$ against $H_1: \theta > 0$ and since the class of power one tests for these hypotheses is contained in the class of power one tests for $H_0: \theta = 0$ against $H_1: \theta > 0$, this test is also asymptotically optimal in the smaller class.
- (2) In a separate paper we show how the exact distribution of the sample size of a test with upper boundary may be computed. This result enables us to construct an asymptotically optimal test with size close to prescribed α (the procedure of Section 3 has size considerably smaller than α). Comparison of exact computations with the asymptotic expression for expected sample size as $\theta \downarrow 0$ show that the asymptotic theory cannot even serve as a crude approximation for the expected sample size unless θ is extremely small. This motivates us to study asymptotic theory as $\alpha \rightarrow 0$. Exact comparisons with an alternative procedure are also given.

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